Comparative Analysis of Multilayer Perceptrons and Convolutional Neural Networks for Fashion Image Classification

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Final Project Proposal

Abstract / Brief Description

This project aims to develop and compare two neural network architectures for clothing image classification: a Multilayer Perceptron (MLP) and a Convolutional Neural Network (CNN).

Using the Fashion-MNIST dataset, consisting of 70,000 grayscale images across ten fashion categories, the study will examine the influence of neural network architecture on classification accuracy, computational efficiency, and generalization performance.

Proposed Methodology

1. Prepare and normalize the Fashion-MNIST dataset using Python.
2. Develop an MLP baseline model using fully connected neural network layers.
3. Implement a CNN using convolutional, pooling, and dense layers for feature extraction.
4. Train and evaluate both models under comparable experimental settings.
5. Analyze accuracy, loss curves, confusion matrices, and training time to compare performance.

Team Responsibilities

Ali Hamza — MLP Development: Implement and train the Multilayer Perceptron model, perform hyperparameter experiments, and evaluate baseline classification performance.

Nabonita Das — CNN Development: Design and train the Convolutional Neural Network, evaluate spatial feature learning, and analyze classification performance.

Joint Contributions: Dataset preparation, model comparison, result visualization, technical report, and final presentation.

Expected Outcomes

We expect the CNN to achieve stronger classification performance than the MLP because of its ability to capture spatial image features.

The final deliverables will include reproducible Python code, trained models, comparison graphs, a technical report, and a 15-minute recorded presentation.

Project Timeline

Date	Planned Activities
Oct 23	Submit project proposal
Oct 24 – Nov 6	Dataset preparation and individual model implementation
Nov 7 – Nov 13	Initial evaluation and progress update
Nov 14 – Dec 4	Model tuning, comparison, and visualizations
Dec 5 – Dec 14	Final report, code, and recorded presentation

References

[1] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms," *arXiv:1708.07747*, 2017.